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Roll No.

220753

5th Sem/ Civil
Subject : Railways, Bridges & Tunnels

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Importance of bridge are _____
- a) To reduce the travel time
 - b) To reduce accidents arising day by day
 - c) To minimize congestion on roads
 - d) All of the above
- Q.2 What is the primary purpose of bearing in a bridge?
- a) To transfer the load to the superstructure
 - b) To transfer the load to the track
 - c) To transfer the load to the embankment
 - d) To transfer the load to the substructure
- Q.3 What are some common methods for ventilating railway tunnels?
- a) Natural ventilation
 - b) Mechanical ventilation
 - c) A combination of natural & mechanical ventilation
 - d) All of the above

- Q.4 A bridge is called culvert if its span is up to
- a) 12m b) 10m
 - c) 8m d) 6m
- Q.5 In case of railways
- a) A detour round the hill is preferred
 - b) A open cut is preferred
 - c) Tunnelling is preferred
 - d) All of the above
- Q.6 The useful life of timber bridge is _____ years
- a) 5-10 b) 10-15
 - c) 15-20 d) 20-25

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Topographical agricultural and industrial maps are used for the study of _____.
- Q.8 The broad gauge is _____ mm.
- Q.9 _____ is a common material for the manufacturing of rails.
- Q.10 Bearing plates are used in _____ sleepers.
- Q.11 Gravel is _____ than stone ballast.
- Q.12 _____ is the art of constructing a tunnel.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Enlist any four requirements of an ideal permanent way.
- Q.14 Give the requirements of an ideal rail. (any five)
- Q.15 What are the salient feature of concrete sleepers?
- Q.16 Give point of distinction between permanent and temporary bridges.
- Q.17 What do you understand by maintenance of bridges?
- Q.18 What are the function of tunnel? (any five)
- Q.19 Write a short note on lighting of tunnels.
- Q.20 What do you understand by mucking?
- Q.21 Enlist the function of fixtures and fastenings?
- Q.22 Describe the objective of signaling?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 What are the function of railway sleeper. Give any four requirement of a good sleeper. Name various types of sleepers. Explain any one?

- Q.24 Give the classification of tunnels according to its shapes along with diagram?
- Q.25 Write down the various purpose of maintenance and special maintenance?